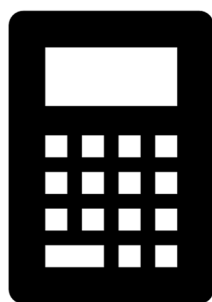


M090: Introductory Algebra

Notes Packet



Fall 2026

Sections 901-927

Table of Contents

Unit 1	1
Section 1.0: The Language of Algebra and Arithmetic Review	3
Section 1.1: Evaluating and Translating Algebraic Expressions	6
Section 1.2: Simplifying Monomial Expressions with Exponents	8
Section 1.3: Adding and Subtracting Polynomials; Distributive Property	12
Section 1.4: Multiplying Polynomials	14
Section 1.5: Adding and Subtracting Rational Expressions	16
Section 1.6: Solving Linear Equations	18
Section 1.7: Solving Linear Equations with Fractions	20
Section 1.8: Solving Formulas for a Specified Variable	23
Section 1.9: Solving Linear Inequalities	25
Unit 2	27
Section 2.0: Intro to Graphing	29
Section 2.1: Linear Equations in Two Variables	32
Section 2.2: Slope of a Line	38
Section 2.3: Finding the Equation of a Line	41
Section 2.4 Intro to Functions	45
Section 2.5 Function Notation	47
Section 2.6 Linear Functions	51
Section 2.7 Applications of Linear Functions	53

Unit 3	57
Section 3.0: Intro to Quadratic Functions	59
Section 3.1: Finding the Vertex and y -intercept for Quadratic Functions	63
Section 3.2: Finding Intercepts of Quadratic Functions (Square Root Property)	66
Section 3.3: Finding Intercepts of Quadratic Functions (Greatest Common Factor)	70
Section 3.4 Part 1: Finding Intercepts of Quadratic Functions (Factoring Trinomials)	74
Section 3.4 Part 2: Finding Intercepts of Quadratic Functions (Factoring Trinomials)	76
Section 3.5: Finding Intercepts of Quadratic Functions (Quadratic Formula)	79
Section 3.6: Finding Intercepts and Vertex for Quadratic Functions	83
Section 3.7: Graphing Quadratic Functions	87

Unit 1

Section 1.0: The Language of Algebra and Arithmetic Review

Key Terms

Algebra: The branch of mathematics that uses variables (letters) to represent numbers in expressions, equations and formulas. We use algebra because it helps us create generalized formulas. For example, we know that the formula for distance traveled is to multiply the rate of speed (r) by time (t) ($D = rt$).

Variable: A letter that represents a number whose value may change (vary)

Constant: A number whose value always stays the same (constant)

Expression: A number, a variable, or a combination of those

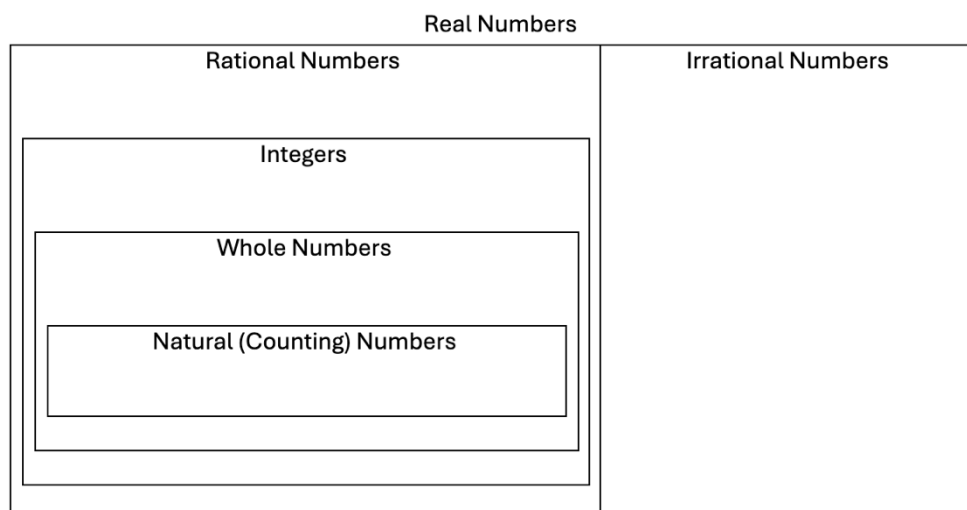
Term/s: A constant, or the product (multiplication) of a constant and one or more variables
When you have more than one term, they are separated by either addition or subtraction.

Factors: The multipliers in a term

Coefficient: The numerical quantity multiplied by the variable in an algebraic expression

Equation: Two expressions connected by an equal sign

Inequality: To show one expression is less than or greater than another expression ($<$, $>$, $\leq$, $\geq$)



Board Work: Simplify each expression into a single fraction or integer value

1. $\frac{1}{8} + \frac{1}{6}$

2. $\frac{3}{5} + \frac{1}{4}$

3. $\frac{3}{4} - \frac{5}{12}$

4. $\left(\frac{4}{7} - \frac{2}{3}\right)^2$

5. $\frac{1}{9} \cdot \frac{1}{6}$

6. $(4)\left(\frac{6}{7}\right)$

7. $\frac{3}{8} \div \frac{1}{16}$

8. $\frac{\frac{5}{8}}{\frac{3}{4}}$

9. $\frac{16 - 3}{4 + 9}$

10. $\frac{(16)(3)}{(9)(4)}$

11. $\frac{16}{0}$

12. $\frac{0}{16}$

13. $24 \div 4 \cdot 2$

14. $17 - 5 + 8$

Section 1.1: Evaluating and Translating Algebraic Expressions

Evaluating Algebraic Expressions

In math, to **evaluate** means to find the **numerical** value of an expression when the variable(s) are replaced by given numbers.

1. Always use parentheses to substitute any variables with numbers.
2. Simplify the expression completely according to the correct order of operations.

Ex 1: Evaluate given $d = -2$ and $f = 5$:

$$\frac{d^2 - f^2}{d^2 + f^2}$$

Ex 2: If $x = -1$, $y = -3$, and $z = 2$, find:

$$\frac{2x^2 + y^2}{|-10 + z^3|}$$

Ex 3: For $a = 5$, $b = -6$, and $c = -3$, evaluate:

$$b^2 - 4ac$$

Ex 4: Given $a = 1$, $b = 7$, and $c = -6$, find:

$$\frac{-b + \sqrt{b^2 - 4ac}}{2a}$$

Ex 5: Evaluate for $a = 2$, $b = -10$, and $c = 8$:

$$\frac{-b + \sqrt{b^2 - 4ac}}{2a}$$

Translating between English Phrases and Algebraic Expressions

English Phrase	Algebraic Expression
the sum of a number and five	
six less than three times a number	
the quotient of two unknown numbers	
seven more than nine times a number	
the square of the sum of a number and forty-two	
a number decreased by the product of eleven and that same number	
the difference between a number cubed and the opposite of that number	
the product of three numbers	
five times a number reduced by the square of a second number	
three-fourths of a number increased by one hundred	
fourteen less than half of a number	
two-thirds of a number	
the positive difference between two numbers	
the square of the difference between a number and eight	
	$6a^3 + 13$
	$5b^2 - 12$
	$\frac{2m}{n^2}$

Section 1.2: Simplifying Monomial Expressions with Exponents

Exponent Properties for Monomial Expressions

RULE	SYMBOLISM	EXAMPLE	DEFINITION
ZERO EXPONENT	$a^0 = 1$		Any nonzero number raised to the power of 0 is 1
PRODUCT RULE	$a^m \cdot a^n = a^{m+n}$		When multiplying , if the bases are the same, ADD the exponents (but KEEP the bases).
QUOTIENT RULE	$\frac{a^m}{a^n} = a^{m-n}$		When dividing , if the bases are the same, SUBTRACT the exponents (but KEEP the bases)
POWER TO A POWER	$(a^m)^n = a^{mn}$		When raising a power to a power , MULTIPLY the exponents.
NEGATIVE EXPONENT	$a^{-m} = \frac{1}{a^m}$		A negative exponent means to apply the reciprocal property.

Example 1: Simplify each expression as much as possible.

A. t^0

B. $8^0 \cdot x^0 \cdot z^0$

C. $8 \cdot x^0 \cdot z^0$

D. $\left(\frac{3ab^5}{6a^3b^9}\right)^0$

Example 2: Use the properties of exponents to fully simplify each expression.

A. $a^2 \cdot a \cdot a^5$

B. $3b^3 \cdot -2c^8 \cdot 4b^2 \cdot c$

C. $(2xy^2)(3xy^3)$

D. $(-x^6y^2)(3x^4y)$

Example 3: Fully simplify the given expression using only positive integer exponents.

A. $\frac{r^{10}}{r^2}$

B. $\frac{2m^2}{5n^3} \cdot \frac{10n^5}{m^3}$

C. $\frac{2m^2}{5n^3} \div \frac{10n^5}{m^3}$

D. $\frac{8x^9y^8z^5}{12x^4y^{12}z^0}$

Example 4: Simplify using only positive integer exponents.

A. $12x^{-4}$

B. $3a^{-2}b^9$

C. $\left(\frac{5}{8}\right)^{-2}$

D. $(4y)^{-3}$

E. $\frac{n^{10}}{p^{-10}}$

F. $\frac{15r^{-4}s^8}{3rs^3}$

Example 5: Simplify fully.

A. $(5x^4)^2$

B. $(-2x^3y^7)^3$

C. $(-a^9b^6)^5$

D. $\frac{w^3 \cdot w^9}{(w^5)^4}$

E. $\frac{\left(\frac{2x^6w^5}{3xw^5}\right)}{\left(\frac{7x^5w^2}{9x^2}\right)}$

F. $\frac{\left(\frac{3x^{11}y^5}{2x^{-1}y^8}\right)}{\left(\frac{2x^5y^{12}}{5x}\right)}$

Section 1.3: Adding and Subtracting Polynomials; Distributive Property

Key Terms

A monomial, or two or more monomials combined by addition or subtraction, is a **polynomial**. Some polynomials have special names, based on the number of terms.

A **monomial** is a polynomial with exactly one term.

A **binomial** has exactly two terms.

A **trinomial** has exactly three terms.

There are no special names for polynomials with more than three terms.

Like Terms: When two terms are both constants OR have the exact same variable(s) and exponent(s), we say they are **like terms**. Only the like terms can be *combined* through addition or subtraction. To do so, **add or subtract only the coefficients**. The variables never change through addition or subtraction.

Example 1: Simplify the polynomial by combining like terms.

A. $6y + 12 - 7y$

B. $9m^2 - 24m + 12m - 32$

C. $3n^2 - 17n - n + 4n^2$

D. $5x^3 + 3x^2 - 2x - 9 - 12x^3 - 10x + 4$

Example 2: Perform the indicated operation and simplify.

A. $(5x^3 + 2x^2 - 13x) + (4x^3 - 10x^2 + 9)$

B. $(2y^6 + y^4 - y) - (8y^6 - 3y^4 + 9y)$

The **distributive property** of multiplication over addition or subtraction allows us to simplify more complicated expressions involving parentheses. For example, $2(5x^2 - 3x + 9)$ can be written as $10x^2 - 6x + 18$ by multiplying 2 by each term inside the parentheses.

Example 3: Perform the indicated operations and simplify fully.

A. $-2x(3x^2 + 8x - 4)$

B. $2(8y - 5) + 7(-2y - 6)$

C. $-x(x^2 - 4) - 2(x^3 + 2x)$

D. $y(7y^2 + 4y - 9) - 3(8y^3 - 3y^2 - 4y)$

E. $3 + 2(5x - 4(x - 9)) + 7x$

F. $11x - 8(y + 5(y - 2(y - x)))$

Section 1.4: Multiplying Polynomials

Example 1: Use the distributive property to multiply and fully simplify each of the following.

A. $(2x + 4)(3x + 5)$

B. $(x - 4)(5x - 1)$

C. $2(b - 3)(b + 4)$

D. $4(w + 2)(w - 2)$

E. $(2y - 7)^2$

F. $2(y - 7)^2$

H. $(w + 2)(w^2 - 2w + 9)$

I. $(2x - 1)(x^2 + 3x - 10)$

Example 2: Find the **perimeter** of a rectangle if the length is $3x - 4$ and the width is $2x + 7$.

Example 3: Find the **area** of a rectangle if the length is $3x - 4$ and the width is $2x + 7$.

Example 4: A given rectangle has a length of $2x + 3$ and a width of $x - 6$. If the length is decreased by 7 and the width is increased by 12, find the area of the new rectangle in terms of x .

Section 1.5: Adding and Subtracting Rational Expressions

Example 1: Find the sum or difference.

A. $\frac{1}{6} + \frac{3}{4}$

B. $\frac{7}{y} - \frac{x}{y}$

Example 2: Add or subtract the rational expressions by obtaining a common denominator.

A. $\frac{5}{w} - \frac{7}{2w}$

B. $\frac{1}{6x} + \frac{2}{9x}$

C. $\frac{3}{y} + \frac{5}{x}$

D. $\frac{11y}{2x} - \frac{x}{5y}$

E. $\frac{2}{7r} + \frac{3}{7}$

F. $\frac{4}{5} - \frac{9}{5w}$

Example 3: Add or subtract the expressions. Simplify if possible.

A. $\frac{3}{7x^2} + \frac{5}{14x}$

B. $\frac{2}{x} - \frac{9}{10x^4} + \frac{3}{x}$

C. $\frac{2}{3b} + \frac{1}{4} + \frac{1}{6b}$

D. $\frac{2}{5r} - \frac{3}{4r} - \frac{7}{10}$

E. $\frac{2x - 5}{3x} - \frac{5x + 9}{6x}$

F. $\frac{3y + 7}{8y} - \frac{9y - 11}{4y}$

Section 1.6: Solving Linear Equations

Example 1: Solve the linear equation for the given variable.

A. $3x - 8 = -12$

B. $6 - 4w = 15$

C. $y + 4 + 5y - 9 = 87$

D. $6r - 3r + 94 + 2r = 18$

Example 2: Solve each equation. Write your answer in simplest fraction form.

A. $10(a - 5) + 2 = -3$

B. $6(x - 2) - 8 = 3x - 20$

C. $2(7 - x) + 11 = 12 - 2x$

D. $-4 - (3 - x) - 2 = 9(x - 1) - 8x$

Classifying linear equations

- An equation that has a finite number of unique real solutions is called **conditional**.
- An equation that has **no solution** is sometimes referred to as a **contradiction**.
- An equation that has **infinitely many solutions (all real numbers)** is called an **identity**.

Example 3: Go back to Examples 1 and 2 and classify each as **conditional**, **contradiction**, or **identity** based on the solution(s) you found.

In Class/Extra Practice:

1. $3x - 2 = -9x + 15$

2. $r - 3 - 7 + r = 10 + 2r$

3. $3(w - 4) = 3(w - 9) + 15$

4. $-4x + 4(x - 4) = -11$

5. $-2(r + 3) + 7 = 1 - 2r$

6. $-(-y - 28) = 7(y + 4)$

Section 1.7: Solving Linear Equations with Fractions

Reminder about equations: We can do any operation we want to one side of an equation, so long as we do it to the other side of the equation.

Example 1: Solve the equation by dividing both sides by the coefficient.

$$\frac{3}{4}x = 15$$

Example 2: Now we will solve the same equation by the method of **clearing fractions**. With this method, we will multiply both sides of the equation by a number (LCD) that will eliminate all the fractions. Let's try it!

$$\frac{3}{4}x = 15$$

Did you know that

$$\frac{3}{4}x = \frac{3x}{4}$$

Sometimes it is
easier to write it
one way vs the
other

Example 3: For the following examples, use the method of clearing fractions to solve.

A. $\frac{x}{-5} = 10$

B. $\frac{1}{2}y + \frac{3}{4} = 10$

C. $\frac{1}{2}x - \frac{2}{7} = \frac{5}{4} - x$

D. $\frac{35 + x}{5} = 18$

E. $\frac{1}{7}(x - 4) + 5 = \frac{1}{2}(3x + 4)$

F. $x - \frac{7}{8} = \frac{5}{6}$

G. $\frac{1}{3}(3m + 1) = \frac{1}{4}(m - 4) + 1$

Section 1.8: Solving Formulas for a Specified Variable

Equations which contain multiple variables are often referred to as *literal equations* or *formulas*.

The process of isolating a specific variable in the equation or formula is known as *solving* for the specified variable.

Example 1: Solve each formula for y .

A. $3x + y = 8$

B. $2x - 5y = 12$

C. $Z = x + xwy$

D. $5xy = 19$

Example 2: Solve each literal equation for the specified variable.

A. $a + b - c = d$; solve for b

B. $ax^2 + bx + c = 0$; solve for b

C. $m = \frac{r - 7n}{5}$; solve for r

D. $t = \frac{d}{r}$; solve for r

E. $P = 2L + 2W$; solve for W

F. $V = LWH$; solve for W

REVIEW: Solve the following equations by clearing fractions.

A. $y - \frac{2}{5} = \frac{3}{2}y + \frac{4}{5}$

B. $\frac{2}{7} = \frac{3}{7}(x - 4) - \frac{5}{7}$

Section 1.9: Solving Linear Inequalities

Example 1: Represent each inequality on a number line graph. Use the graph to convert to interval notation.

A. $x > 5$

B. $x < -6$

C. $-3 \leq x$

D. $12 \geq x$

E. $-3 < x < 1$

F. $2 \leq x < 3$

G. $x < -7$ or $x \geq 0$

H. $x \leq 2$ or $x > 5$

The process of solving linear inequalities is similar to solving linear equations, with one major exception:

In order for the inequality to remain a true statement, the direction of the inequality symbol must be **reversed** upon **dividing or multiplying** by a **negative** number.

Example 2: Solve each inequality. Write your final answer in interval notation.

A. $-6 + 3x \leq -12 + 5x$

B. $4(5x - 7) > 10 + 5x$

C. $\frac{4}{5}y + 5 < 6$

D. $2 \leq \frac{x}{-8} - 3$

E. $\frac{x - 30}{-4} \geq 50$

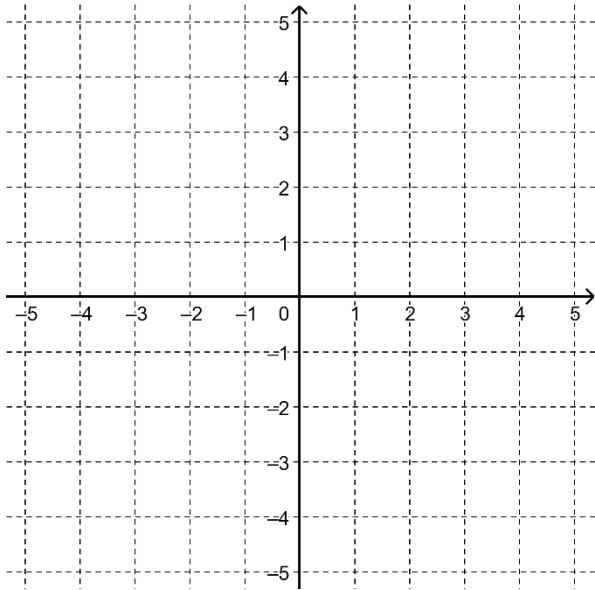
F. $\frac{x - 30}{4} > 50$

Unit 2

Section 2.0: Intro to Graphing

Plotting Solutions to an Equation in the Rectangular Coordinate System

We can represent an ordered pair (x, y) using the coordinate (Cartesian) plane.



Example 1: Plot and label the following ordered pairs in the rectangular (Cartesian) coordinate plane.

- A. $(4, 0)$
- B. $(-1, 3)$
- C. $(0, 5)$
- D. $(-3, -4)$

Definition: Solution of an Equation Involving x and y

An ordered pair (x, y) is a solution to an equation involving x and y if the equation is a true statement when the x - and y - values of the ordered pair are substituted into the equation.

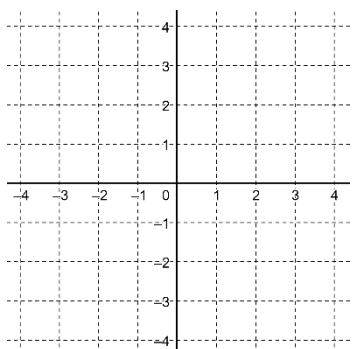
Basic Graphing Strategy

When graphing, x -coordinates are the input values and y -coordinates are the output values. The graph is the set of all such (x, y) points. Hence, when graphing, choose x -values and use the algebraic expression of the equation to calculate corresponding y -values. Plot enough points to be able to determine the overall shape of the graph.

Graphs of Basic Equations

Graph the following equations by plotting points. Use interval notation to state the domain (x -values) and range (y -values) of each equation.

A. $y = 3$ (Constant/Polynomial)

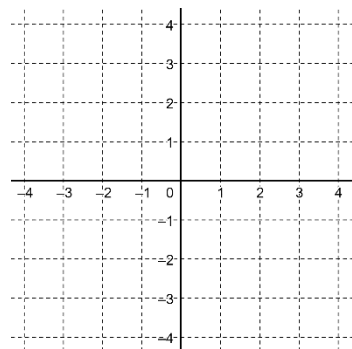


x	y

Domain:

Range:

B. $y = x$ (Linear/Polynomial)

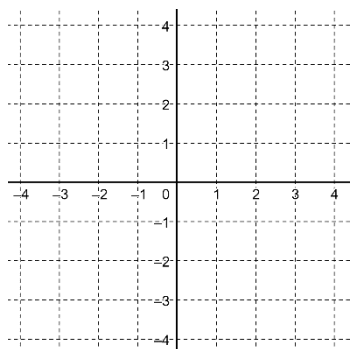


x	y

Domain:

Range:

C. $y = x^2$ (Quadratic/Polynomial)

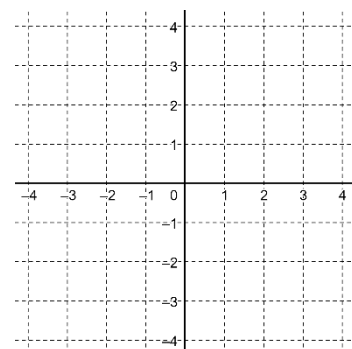


x	y

Domain:

Range:

D. $y = \sqrt{x}$ (Square Root)

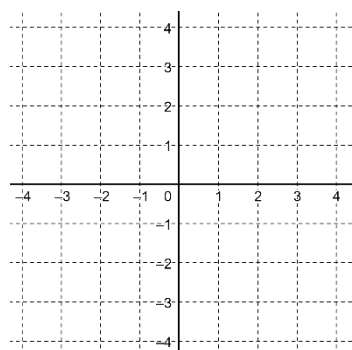


x	y

Domain:

Range:

E. $y = |x|$ (Absolute Value)

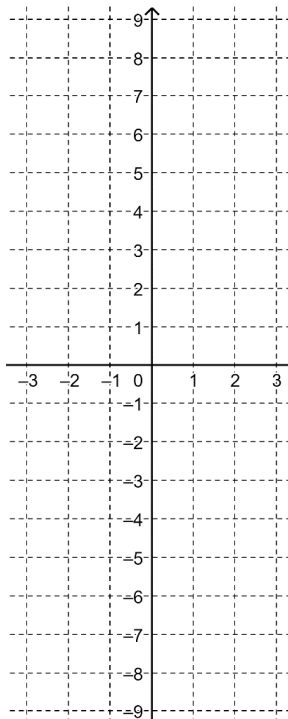


x	y

Domain:

Range:

F. $y = x^3$ (Cubic/Polynomial)



x	y

Domain:

Range:

G. $y = \sqrt[3]{x}$ (Cube Root)



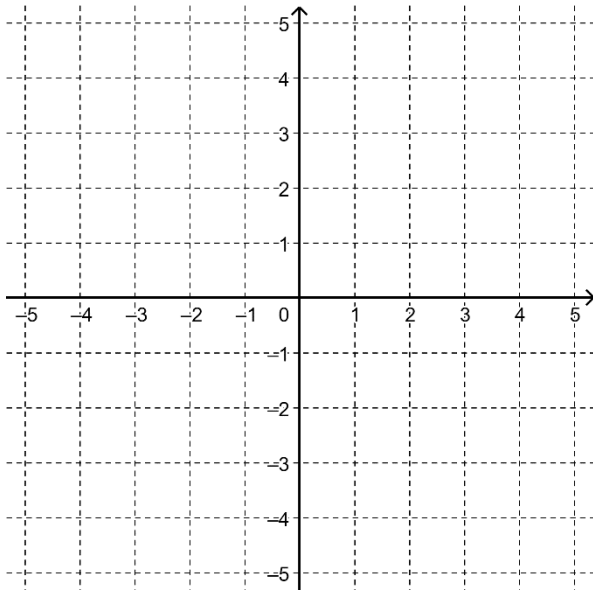
x	y

Domain:

Range:

Section 2.1: Linear Equations in Two Variables

Example 1: Graph the following linear equation using a table. $2x + 3y = 6$



x	y
0	
	0

Now solve the equation for y .

$$2x + 3y = 6$$

Use the graph to identify the following.

x -intercept:

y -intercept:

Slope:

Slope Intercept Form of a Line

$$y = mx + b$$

where m is the slope of the line and $(0, b)$ is the y -intercept

Example 2: Identify the slope and y -intercept for the following.

A. $y = \frac{1}{4}x - 6$

B. $2x - y = 7$

Definition of Intercepts

x -intercept: Any point on a graph with a y -value of 0.

y -intercept: Any point on a graph with an x -value of 0.

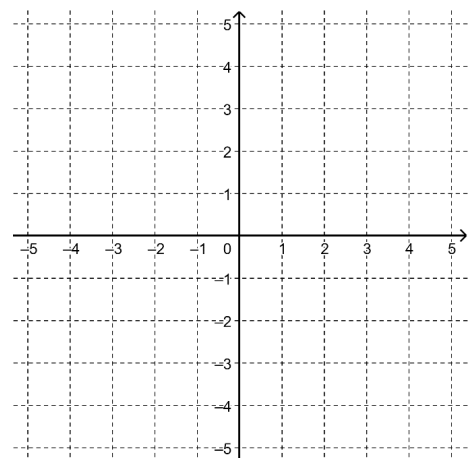
Example 3: Find the x and y -intercepts algebraically for the following. Then re-write the equation into slope-intercept form.

A. $5x + 2y = 6$

x -intercept

y -intercept

Slope-intercept form

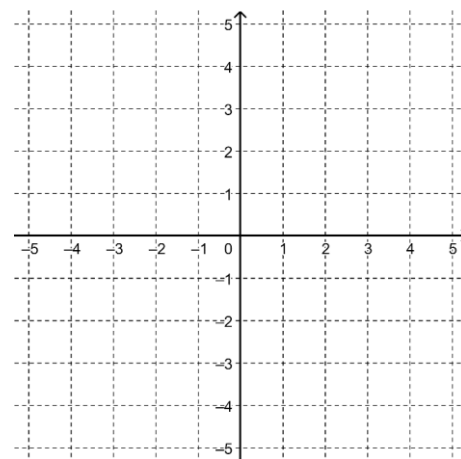


B. $4x - 5y = 10$

x -intercept

y -intercept

Slope-intercept form

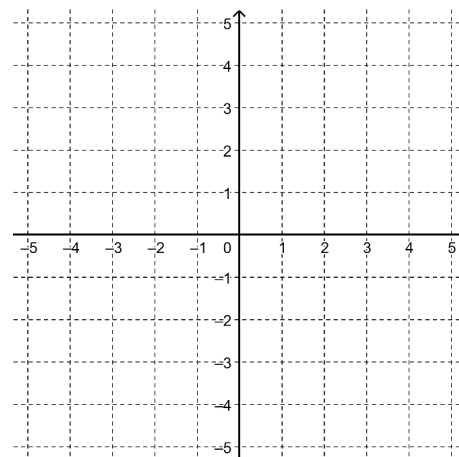


C. $y = -\frac{3}{2}x - 3$

x-intercept

y-intercept

Slope-intercept form

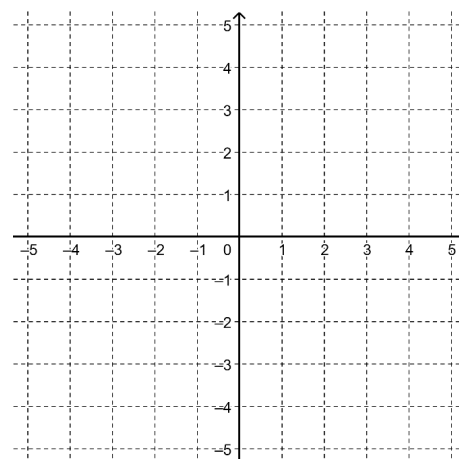


D. $y = 2x$

x-intercept

y-intercept

Slope-intercept form



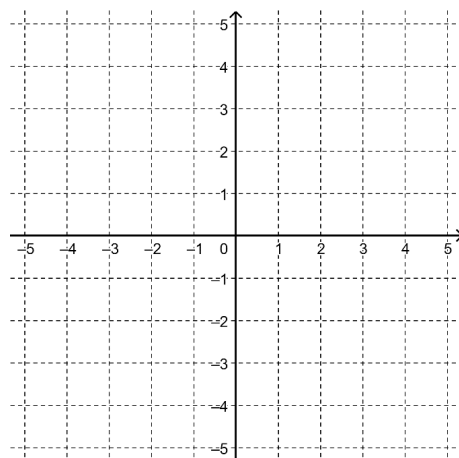
Horizontal and Vertical Lines

E. $y = -2$

x -intercept

y -intercept

Slope-intercept form

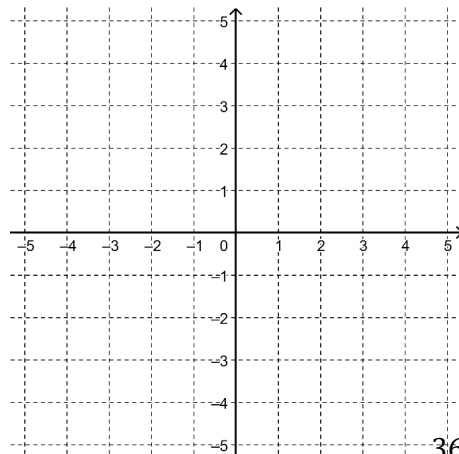


F. $x = 3$

x -intercept

y -intercept

Note: Equations of vertical lines cannot be written in slope-intercept form.



Section 2.2: Slope of a Line

Slope Between Two Points on a Line

$$\text{Slope} = m = \frac{\text{rise}}{\text{run}} = \frac{\text{change in } y}{\text{change in } x} = \frac{y_2 - y_1}{x_2 - x_1}$$

where (x_1, y_1) and (x_2, y_2) are two points on the line

Example 1: Find the slope of the line that passes through the two given points.

A. $(2, 3), (5, 7)$

B. $(3, -4), (-2, -8)$

C. $(3, 7), (3, -10)$

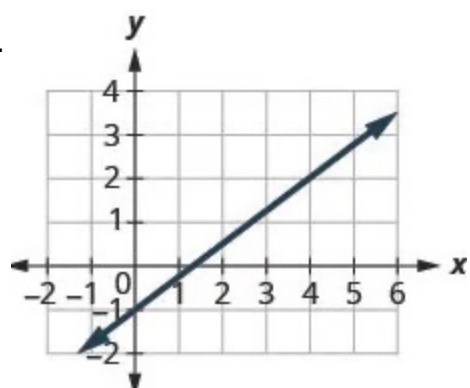
D. $(-2, -5), (3, -5)$

E. $\left(\frac{4}{3}, \frac{1}{3}\right), \left(-\frac{3}{5}, \frac{7}{5}\right)$

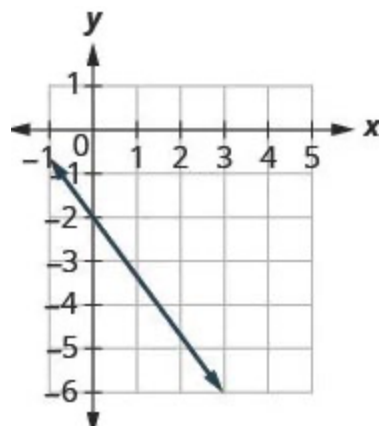
F. $\left(\frac{1}{7}, \frac{2}{5}\right), \left(-\frac{4}{5}, \frac{2}{5}\right)$

Example 2: Find the slope of the line shown in the graph.

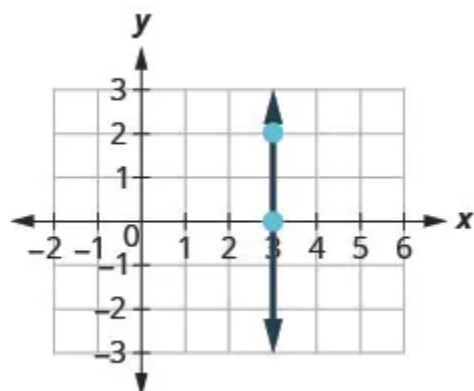
A.



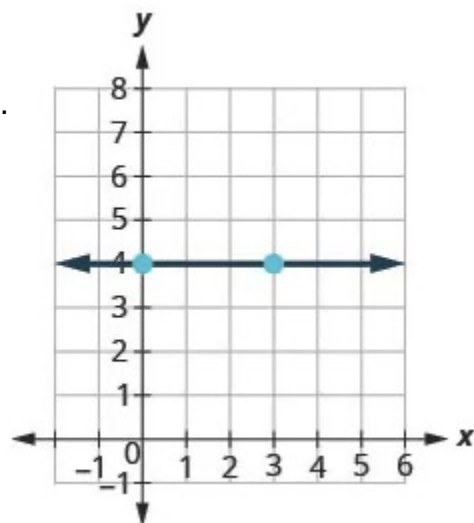
B.



C.



D.

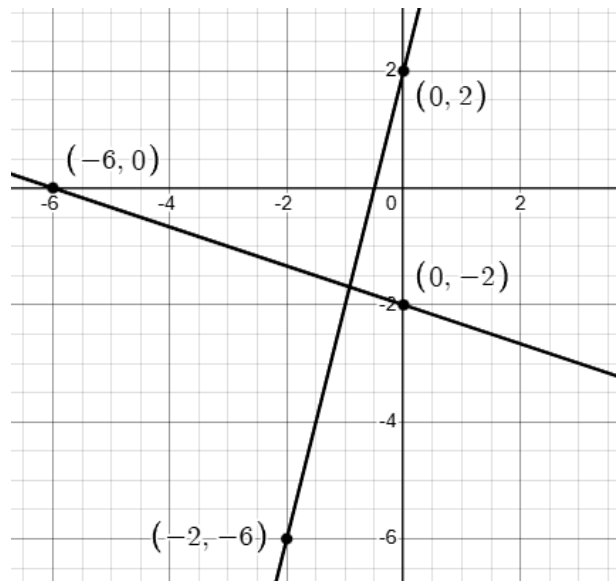


Parallel and Perpendicular Lines

Two lines are said to be **parallel** if they have the same slope (with different y -intercepts)

Two lines are said to be **perpendicular** if their slopes are opposite reciprocals.

Example 3: Determine if the two lines are parallel, perpendicular, or neither.



Example 4: Determine if the two lines are parallel, perpendicular, or neither.

$$\begin{aligned}x + y &= 5 \\ -2x - 2y &= 7\end{aligned}$$

Example 5: Determine if the two lines are parallel, perpendicular, or neither.

$$\begin{aligned}2y &= x - 2 \\ y &= -2x - 4\end{aligned}$$

Section 2.3: Finding the Equation of a Line

Point Slope Form of a Line

$$y - y_1 = m(x - x_1)$$

Where m is the slope and (x_1, y_1) is any point on the line

*Can be used to find the equation of any **non-vertical** line.

Example 1: Find the equation of the line with a y -intercept of $(0, -3)$ and a slope of $m = -\frac{4}{3}$.

Example 2: Find the equation of the line with x -intercept of $(-3, 0)$ and a slope of $m = -\frac{4}{3}$.

Example 3: Find the equation of the line that contains the points $(2, 3)$ and $(-6, 1)$.

Example 4: Find the equation of the line that contains the points $(1,7)$ and $(-3,7)$.

Example 5: Find the equation of the line that contains the points $(2, -8)$ and $(2,1)$.

Example 6: Find the equation of the line containing the point $(-1,3)$ and parallel to the line $2x + y = 10$.

Example 7: Find the equation of the line containing the point $(2, -3)$ and perpendicular to the line $4y - x = 20$.

In Class Practice

1. Find the equation of the line with a y -intercept of $(0,5)$ and a slope of $m = \frac{3}{5}$.

2. Find the equation of the line with x -intercept of $(5,0)$ and a slope of $m = -\frac{3}{5}$.

3. Find the equation of the line that contains the points $(-2, -3)$ and $(6,1)$.

4. Find the equation of the line that contains the points $(2, -9)$ and $(2, -3)$.

5. Find the equation of the line that contains the points $(5, -2)$ and $(4, -2)$.

6. Find the equation of the line containing the point $(2, 1)$ and parallel to the line $3x - y = 7$.

7. Find the equation of the line containing the point $(6, 3)$ and perpendicular to the line $2y - x = 20$.

Section 2.4 Intro to Functions

Relation: Any set of ordered pairs

Input: First value in the ordered pair

Output: Second value in the ordered pair

Domain: The set of all input values

Range: The set of all output values

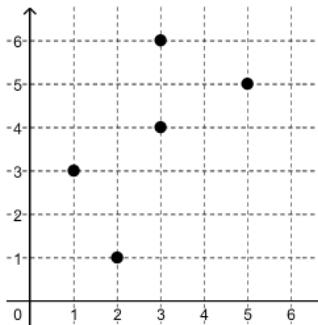
Example 1: For the following relation, determine the domain and range.

$$\{(1, -1), (2, 5), (3, 10), (4, 16), (2, -3)\}$$

Domain:

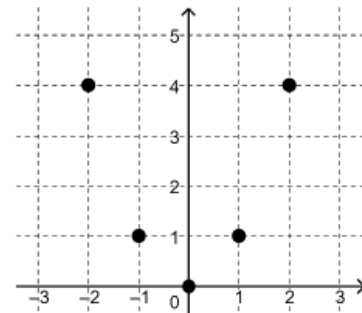
Range:

Example 2: For the following graphs, determine the domain and range.



Domain:

Range:



Domain:

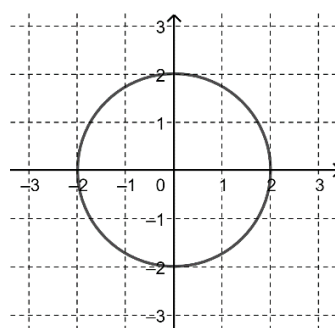
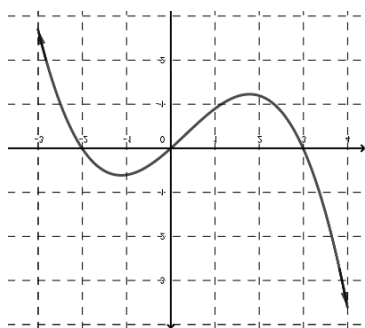
Range:

A **function** is a relation that assigns each element in its domain to exactly one element in its range.

- One input value cannot correspond to two different output values.
- However, one output value can correspond to multiple input values.

Example 3: Based off the definition of a function, determine if Example 1 and Example 2 are functions.

Example 4: Determine if the graphs below represent a function.



Vertical Line Test

A set of points in a rectangular coordinate system is the graph of a function if **every** vertical line intersects the graph in **at most one** point.

If **any** vertical line intersects the graph in **more than one** point, the graph does not represent a function.

Section 2.5 Function Notation

Function Notation

The graph of a function is the graph of all its ordered pairs (x, y) or using function notation, $(x, f(x))$ where $y = f(x)$.

f	name of function
x	x -coordinate of the ordered pair (input)
$f(x)$	y -coordinate of the ordered pair (output)

Example 1: Consider the linear equation $y = 4x - 5$. Find the value of y when $x = 2$.

Example 2: Consider the linear function $f(x) = 4x - 5$. Find $f(2)$.

Function notation is useful in mathematics for several reasons.

- It guarantees that the relation in question is a function, which means each element of the domain corresponds to exactly one element of the range and that its graph passes the vertical line test.
- It provides a useful naming mechanism when working with multiple functions.
- It yields a convenient shorthand (as in Example 2 above) that shows how the output depends on the input.

Example 3: Let $g(x) = -2x + 7$ and $h(x) = 3x - 5$. Evaluate each of the following. Use proper notation with each step.

A. $g(-1)$

B. $h(-1)$

C. $h(0)$

D. $g(0)$

E. $g\left(\frac{5}{2}\right)$

F. $h(a)$

G. $g(x - 7)$

H. $h(k + 1)$

Example 4: Let $f(x) = 5x + 3$. Solve the following for x .

A. $f(x) = 18$

B. $f(x) = -8$

Example 5: Use the graph of the function $y = h(x)$ below to answer each prompt.

A. $h(2) =$ _____

B. $h(x) = -3; x =$ _____

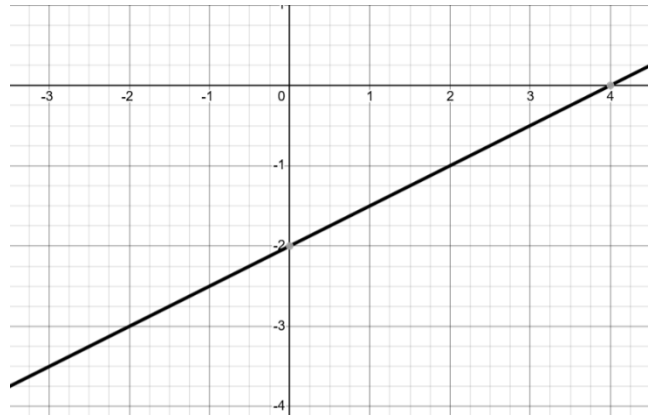
C. $h(4) =$ _____

D. $h(x) = -2$ when $x =$ _____

E. x -intercept: _____

F. y -intercept: _____

G. slope: _____



In Class Practice:

1. Let $f(x) = 2x - 8$ and $g(x) = x - 9$. Evaluate each of the following. Use proper notation with each step.

A. $g(-2)$

B. $f(4)$

C. $g(0)$

D. $g\left(\frac{1}{2}\right)$

2. Let $f(x) = 2x - 8$ and $g(x) = x - 9$. Evaluate each of the following. Use proper notation with each step.

A. $f\left(-\frac{1}{2}\right)$

B. $g(k)$

C. $f(x + 7)$

D. $g(a - 1)$

3. Let $p(x) = 2x - 11$. Solve the following for x .

A. $p(x) = -15$

B. $p(x) = 3$

4. Use the graph of the function $y = f(x)$ to the right to answer each prompt.

A. $f(2) = \underline{\hspace{2cm}}$

B. $f(x) = -3$ when $x = \underline{\hspace{2cm}}$

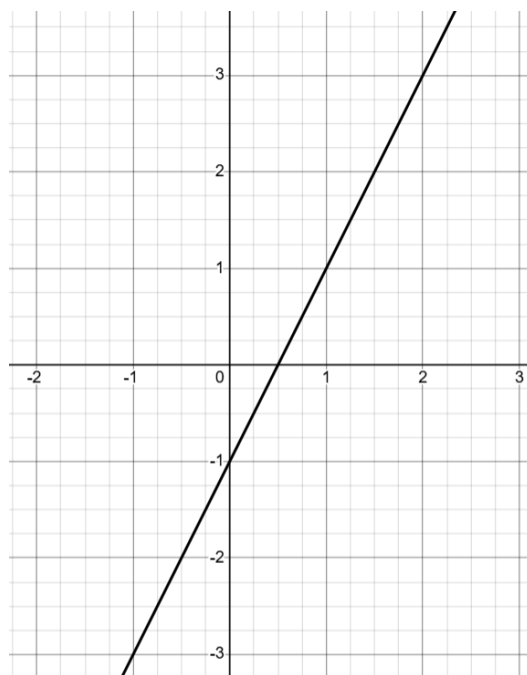
C. $f(1.5) = \underline{\hspace{2cm}}$

D. $f(x) = -1$; $x = \underline{\hspace{2cm}}$

E. x -intercept: $\underline{\hspace{2cm}}$

F. y -intercept: $\underline{\hspace{2cm}}$

G. slope: $\underline{\hspace{2cm}}$



Section 2.6 Linear Functions

Function Notation for Lines

$$f(x) = mx + b$$

where m is the slope of the line and $(0, b)$ is the y -intercept

- When creating a linear function (writing the equation of a non-vertical line)

Creating a Linear Function

- If both the slope and y -intercept are known:
 - Directly substitute your m and b into $f(x) = mx + b$
- If both the slope and/or the y -intercept are unknown:
 - If necessary, find the slope using the formula:
 - Identify an ordered pair (x_1, y_1) .
 - Use point-slope form $y - y_1 = m(x - x_1)$ with your slope and point. Then solve for y .
 - Since it is a function, replace your y with $f(x)$.

Example 1: Graph the function $f(x) = 2x - 4$, and identify the following:

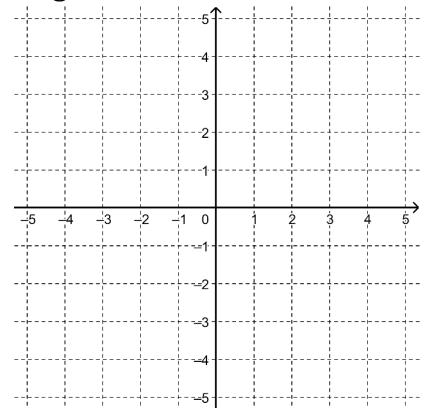
Slope:

y -intercept:

x -intercept:

Domain:

Range:



Example 2: Graph the function $g(x) = -x + 4$, and identify the following:

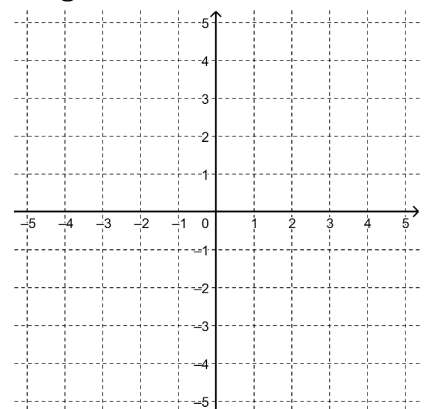
Slope:

y -intercept:

x -intercept:

Domain:

Range:



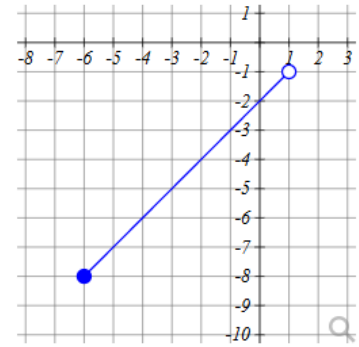
Example 3: Using the graph at the right answer the following questions.

Domain (in interval notation):

Range (in interval notation):

$$f(-2) = \underline{\hspace{2cm}}$$

$$\text{When } f(x) = -4, x = \underline{\hspace{2cm}}$$



Example 4: Find the equation of the linear function $h(x)$ that has a slope of $\frac{3}{4}$ and a y-intercept at $(0, -5)$.

Example 5: Find the equation of the linear function $k(x)$ that has a slope of $\frac{1}{5}$ and passes through the point $(-5, 0)$.

Example 6: Find the equation of the linear function $f(x)$ that passes through the points $(2, 3)$ and $(4, 9)$.

Example 7: Find the equation of the linear function $g(x)$ that contains the points $(7, 9)$ and $(-4, 9)$.

Section 2.7 Applications of Linear Functions

$f(x) = mx + b$ where m is the slope (rate) and b is the y -intercept (initial value)

First decide if you are given the rate and initial value or if you will need to create two data points from the information given.

Example 1: Kim owns a ski rental business in Bozeman. His monthly fixed costs are \$2450. This includes his rent, utilities and supplies. Additionally, Kim has one worker whom he pays \$15 per hour. Write a linear function $C(h)$ that models Kim's monthly cost in terms of h , the number of hours his employee works.

Find the monthly cost if Kim pays an employee to work 25 hours.

Example 2: You purchased your car in 2023 for \$22,500. After researching that particular model you learned that the car would depreciate linearly by \$1500 per year. Write a function $V(t)$ that gives the value of the car t years after 2023.

Example 3: Speaking of buying cars, did you know that a car salesperson earns commission on the sale of a car? The week that you bought your car in 2023 the salesperson sold 3 cars and earned \$760 for the week. The week after that, they sold 5 cars and earned \$920 for the week. Find a linear function $E(x)$ that represents the amount of money earned in dollars when x cars are sold in a week.

Example 4: The systolic blood pressure in adults can be modeled using a linear function. Let x represent the age of an adult in years and $P(x)$ represent the systolic blood pressure in millimeters of mercury (mmHg). A 23 year old adult has a systolic blood pressure of 120 mmHg whereas a 53 year old adult has a systolic blood pressure of 132 mmHg. Find the linear function $P(x)$.

Example 5: The relationship between elevation and the boiling point of water can be modeled with a linear function. The boiling point of water at 0 ft elevation (think about that camping trip on the beach) is 212°F . But, if you were to hike up to the “M” here in Bozeman where the elevation is 5700 ft, you will only need to reach a temperature of 200°F for your water to boil. Write a linear function $B(x)$ that represents the boiling point of water, in degrees Fahrenheit, in terms of your elevation, x , in feet.

Example 6: In 2001 the volume in a grain bin is 45 tons. In 2025, there were only 15 tons remaining. Create a linear model $V(t)$ that represents the volume of grain left in the bin, t years after 2000.

Determine when the grain bin will be empty.

Unit 3

Section 3.0: Intro to Quadratic Functions

Quadratic Function (in the polynomial function family)

- $f(x) = ax^2 + bx + c$, where $a \neq 0$, and b and c are real numbers. (General form)
- $f(x) = a(x - h)^2 + k$ (Standard/Vertex Form)

Example 1: $f(x) = -x^2 + 3x + 8$

Quadratic term _____

Coefficient of quadratic term _____

Linear term _____

Coefficient of linear term _____

Constant term _____

Example 2: $f(x) = 2x^2 - 7$

Quadratic term _____

Coefficient of quadratic term _____

Linear term _____

Coefficient of linear term _____

Constant term _____

Example 3: $f(x) = -2 - 3x + \frac{x^2}{3}$ *Put in Descending order first _____

Quadratic term _____

Coefficient of quadratic term _____

Linear term _____

Coefficient of linear term _____

Constant term _____

Example 4: What type of function is $f(x) = 4x + 7$?

Quadratic term _____

Coefficient of quadratic term _____

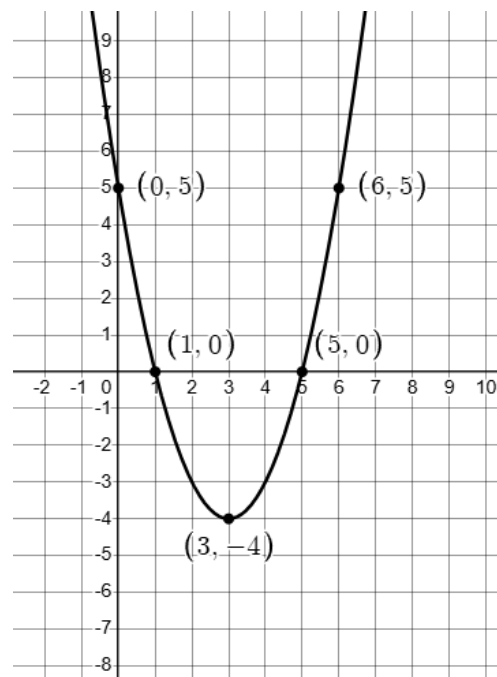
Linear term _____

Coefficient of linear term _____

Constant term _____

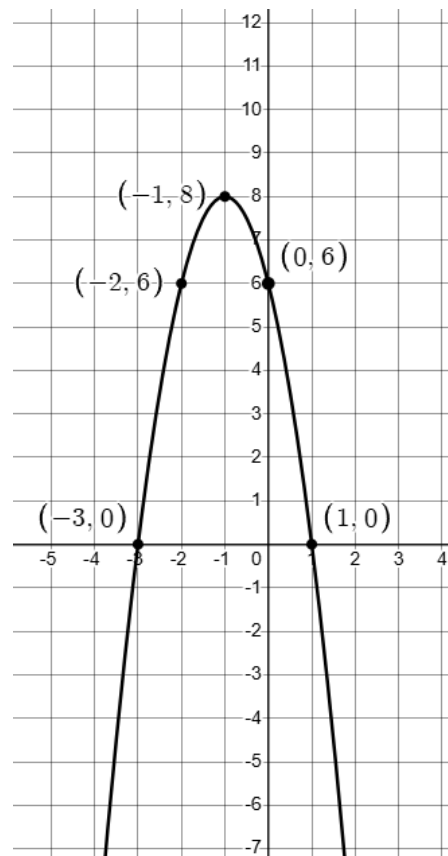
Example 5: Using the graph of the function $f(x) = x^2 - 6x + 5$, find the following:

- A. y-intercept _____
- B. x-intercepts _____
- C. Vertex _____
- D. Axis of Symmetry _____
- E. What is the minimum value of $f(x)$? _____
- F. At what x value does the minimum occur? _____
- G. Domain _____ Range _____



Example 6: Using the graph of the function $f(x) = -2(x + 1)^2 + 8$, find the following:

- A. y-intercept _____
- B. x-intercepts _____
- C. Vertex _____ *Do you see the connection to the function?
- D. Axis of Symmetry _____
- E. What is the maximum value of $f(x)$? _____
- F. At what x value does the maximum occur? _____
- G. Domain _____ Range _____
- H. Simplify the function into general form.



Example 7: Let $g(x) = -x^2 + 2x + 7$ and $h(x) = 3(x - 2)^2 - 5$. Evaluate each of the following. Use proper notation with each step. Simplify completely.

A. $g(-1)$

B. $h(-1)$

C. $h(0)$

D. $g(0)$

E. $h\left(\frac{5}{2}\right)$

F. $g(a)$

G. $g(x - 6)$

H. $h(k + 1)$

Section 3.1: Finding the Vertex and y-intercept for Quadratic Functions

Finding the Vertex and y-intercept

- $f(x) = ax^2 + bx + c$, where $a \neq 0$, and b and c are real numbers. (General form)

The x -coordinate of the vertex: $x_v = \frac{-b}{2a}$

The y -coordinate of the vertex: $f(x_v)$

- $f(x) = a(x - h)^2 + k$ (Standard/Vertex Form)

The x -coordinate of the vertex: $x_v = h$

The y -coordinate of the vertex: $f(x_v) = k$

- Recall: To find the y -intercept we should substitute 0 in for x , which means find $f(0)$.

Example 1: $f(x) = x^2 + 6x - 8$

A. Find the vertex

B. Find the y -intercept

C. Axis of Symmetry

D. Maximum or Minimum Value

Example 2: $f(x) = -\frac{1}{4}x^2 - 3x + 9$

A. Find the vertex

B. Find the y -intercept

C. Axis of Symmetry

D. Maximum or Minimum Value

Example 3: $f(x) = -3(x - 4)^2 + 7$

A. Find the vertex

B. Find the y -intercept

C. Axis of Symmetry

D. Maximum or Minimum Value

Example 4: $f(x) = 5(x - 1)^2 + 3$

A. Find the vertex

B. Find the y -intercept

C. Axis of Symmetry

D. Maximum or Minimum Value

Example 5: $f(x) = -6(x + 2)^2 - 9$

A. Find the vertex

B. Find the y -intercept

C. Axis of Symmetry

D. Maximum or Minimum Value

Example 6: $f(x) = 2x^2 - 7$

A. Find the vertex

B. Find the y -intercept

C. Axis of Symmetry

D. Maximum or Minimum Value

Example 7: Using the graph of the function $f(x) = -x^2 - 4x - 5$

A. y -intercept _____

B. x -intercepts _____

C. Vertex _____

D. Axis of Symmetry _____

E. What is the maximum value of $f(x)$? _____

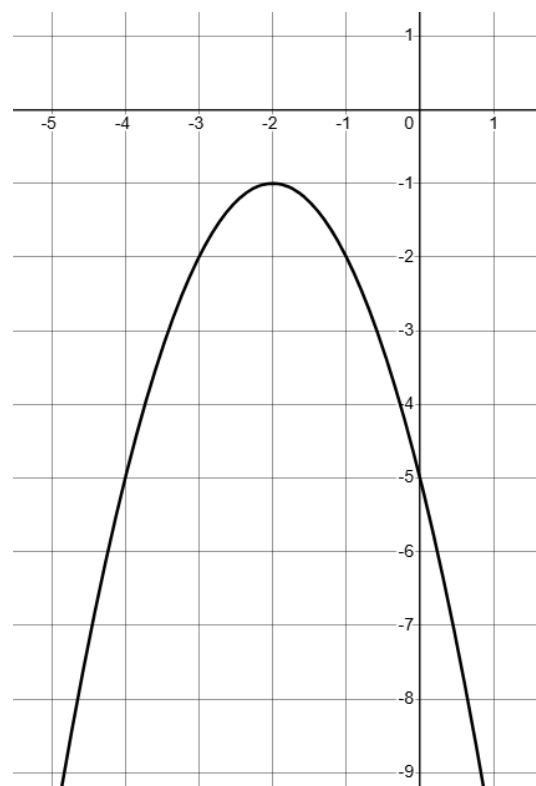
F. At what x value does the maximum occur? _____

G. $f(-4) =$ _____

H. $f(x) = -2$ when $x =$ _____

I. $f(0) =$ _____

J. $f(x) = -5$ when $x =$ _____



Section 3.2: Finding Intercepts of Quadratic Functions (Square Root Property)

If n is a positive number and $x^2 = n$ then, $x = \pm\sqrt{n}$.

If n is a negative number, then $x^2 = n$ has no real number solutions.

If $x^2 = 0$, then $x = 0$.

Example 1: Solve the equation $x^2 = 16$ for x .

]

Example 2: Solve the equation $2x^2 + 2 = 10$ for x .

Recall: To find x -intercepts we substitute in 0 for the y -value. Set $f(x) = 0$.

Example 3: Find the intercepts and vertex for the function $f(x) = x^2 - 36$.

A. Find the x -intercept(s)

B. Find the y -intercept

C. Find the vertex

Example 4: Find the intercepts and vertex for the function $g(x) = 3x^2 - 27$.

A. Find the x -intercept(s)

B. Find the y -intercept

C. Find the vertex

Example 5: Find the intercepts and vertex for the function $f(x) = 10 - 2x^2$.

A. Find the x -intercept(s)

B. Find the y -intercept

C. Find the vertex

Example 6: Find the intercepts and vertex for the function $h(x) = x^2 + 16$.

A. Find the x -intercept(s)

B. Find the y -intercept

C. Find the vertex

Example 7: Find the intercepts and vertex for the function $g(x) = (x - 2)^2 - 9$.

A. Find the x -intercept(s)

B. Find the y -intercept

C. Find the vertex

Example 8: Find the intercepts and vertex for the function $f(x) = -4(x + 3)^2 + 8$.

A. Find the x -intercept(s)

B. Find the y -intercept

C. Find the vertex

Section 3.3: Finding Intercepts of Quadratic Functions (Greatest Common Factor)

Zero Product Property

If the product of two factors is equal to 0, then at least one of the factors must be 0.

If $ab = 0$, then $a = 0$ or $b = 0$

Example 1: Solve the equation $x(x + 3) = 0$.

Factoring out a Greatest Common Factor (GCF)

1. Start by finding the largest numerical value that divides evenly into each term.
2. Determine if every term has a common variable. If so, use the one with the smallest exponent.

Example: $2x^3 + 8x^2$ has a GCF of $2x^2$

Note that factoring is the inverse operation of simplifying: $2x^2(x + 4) =$

Example 2: Solve the equation $2x^2 - 6x = 0$.

Example 3: Solve the equation $4x^2 = -12x$

Example 4: Find the intercepts and vertex for $f(x) = x^2 - 36x$.

A. Find the x -intercept(s)

B. Find the y -intercept

C. Find the vertex

Example 5: Find the intercepts and vertex for $g(x) = 3x^2 - 27x$.

A. Find the x -intercept(s)

B. Find the y -intercept

C. Find the vertex

Example 6: Find the intercepts and vertex for $h(x) = 10x - 2x^2$.

A. Find the x -intercept(s)

B. Find the y -intercept

C. Find the vertex

Example 7: Find the intercepts and vertex for $f(x) = x^2 + x$.

A. Find the x -intercept(s)

B. Find the y -intercept

C. Find the vertex

Example 8: Find the intercepts and vertex for $g(x) = x^2 - 5$.

A. Find the x -intercept(s)

B. Find the y -intercept

C. Find the vertex

Example 9: Find the intercepts and vertex for $f(x) = 3x + 5$.

A. Find the x -intercept(s)

B. Find the y -intercept

Example 10: Find the intercepts and vertex for $g(x) = -(x - 3)^2 + 7$.

A. Find the x -intercept(s)

B. Find the y -intercept

C. Find the vertex

Section 3.4 Part 1: Finding Intercepts of Quadratic Functions (Factoring Trinomials)

Factoring a trinomial with an a value of 1

$$\text{Reminder: } f(x) = ax^2 + bx + c$$

If $a = 1$ then the trinomial **may** factor to:

$$f(x) = (x + d)(x + e) \text{ where } de = c \text{ and } d + e = b$$

Example 1: Find the missing value(s) to make the statement true:

A. $x^2 + 10x + 24 = (x + 4)(x + ?)$

B. $x^2 - 15x + 36 = (x - ?)(x - 12)$

C. $x^2 - 5x + 6 = (x \quad)(x \quad)$

D. $x^2 + 7x - 18 = (x \quad)(x \quad)$

Example 2: Factor the trinomial into the product of two binomials. Be sure to distribute to check your work.

A. $x^2 + 8x + 15$

B. $x^2 - x - 20$

C. $x^2 + 4x + 12$

D. $x^2 + 9x - 10$

E. $x^2 - 11x - 28$

F. $x^2 - 7x + 12$

Recall the Zero Product Property: If $ab = 0$, then $a = 0$ or $b = 0$

Example 3: Solve the following quadratic equations.

A. $x^2 + 3x - 40 = 0$

B. $x^2 - 15x = -44$

C. $x^2 - 60 = 7x$

D. $9x + 8 = -x^2$

Section 3.4 Part 2: Finding Intercepts of Quadratic Functions (Factoring Trinomials)

Example 1: Find the intercepts and vertex for the function $f(x) = x^2 - 5x - 6$.

A. Find the x -intercept(s)

B. Find the y -intercept

C. Find the vertex

Example 2: Find the intercepts and vertex for the function $f(x) = x^2 + 8x - 48$.

A. Find the x -intercept(s)

B. Find the y -intercept

C. Find the vertex

Example 3: Find the intercepts and vertex for the function $h(x) = x^2 + 2x + 1$.

A. Find the x -intercept(s)

B. Find the y -intercept

C. Find the vertex

Example 4: Find the intercepts and vertex for the function $g(x) = x^2 - 17x + 72$.

A. Find the x -intercept(s)

B. Find the y -intercept

C. Find the vertex

Example 5: Use the function $f(x) = x^2 + 4x + 7$ to find and simplify the following.

A. $f(4)$

B. $f(n)$

C. $f(x - 9)$

D. Solve $f(x) = 12$ for x

Section 3.5: Finding Intercepts of Quadratic Functions (Quadratic Formula)

The quadratic formula can be used to solve ANY quadratic equation as long as it is equal to zero and in general form. If $ax^2 + bx + c = 0$, then:

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Example 1: Find the intercepts and vertex for the following functions.

A. $g(x) = x^2 - 17x + 72$

x-intercept(s)

y-intercept

Vertex

B. $h(x) = x^2 - 5x - 7$

x -intercept(s)

y -intercept

Vertex

C. $h(x) = 13x - x^2 + 1$

x -intercept(s)

y -intercept

Vertex

D. $f(x) = -x^2 - 5x + 7$

x -intercept(s)

y -intercept

Vertex

E. $f(x) = -1 + 2x^2 - 3x$

x -intercept(s)

y -intercept

Vertex

Example 2: Use the function $f(x) = 3x^2 - 5x + 7$ to find the following:

A. $f(4)$

B. $f(p)$

C. $f(x + 3)$

D. Solve $f(x) = 13$ for x

Section 3.6: Finding Intercepts and Vertex for Quadratic Functions

Example 1: Find the x -intercept(s).

A. $g(x) = x^2 + 6x$

B. $h(x) = -2x^2 + 5x + 6$

C. $f(x) = 4x^2 - 28$

D. $f(x) = x^2 - 15x + 50$

Example 2: Find the y -intercept.

A. $g(x) = x^2 + 6x$

B. $h(x) = -2x^2 + 5x + 6$

C. $f(x) = 4x^2 - 28$

D. $f(x) = x^2 - 15x + 50$

Example 3: Find the vertex.

A. $g(x) = x^2 + 6x$

B. $h(x) = -2x^2 + 5x + 6$

C. $f(x) = 4x^2 - 28$

D. $f(x) = x^2 - 15x + 50$

Example 4: Use the function $f(x) = -2(x - 15)^2 + 50$ to find the following:

A. x -intercept(s)

B. y -intercept

C. Vertex

D. $f(20)$

Summary of finding intercepts

Section 3.7: Graphing Quadratic Functions

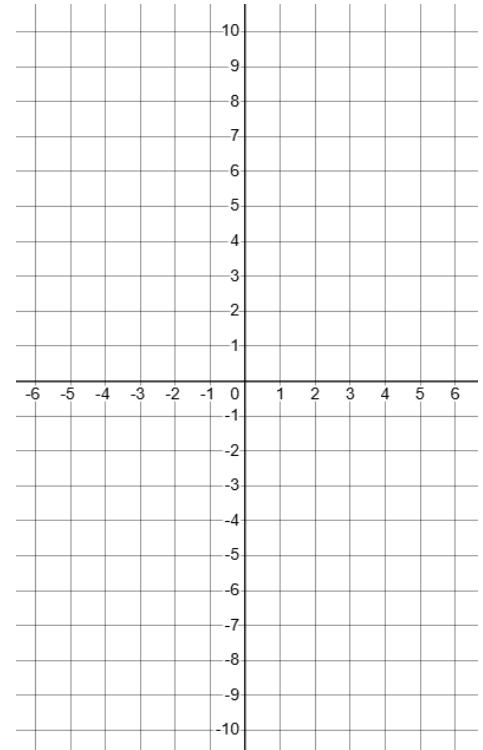
Example 1: Graph the function by first finding the vertex and then completing the table.

$$g(x) = x^2 + 6x + 5$$

Vertex

Table

x	$g(x)$



Domain:

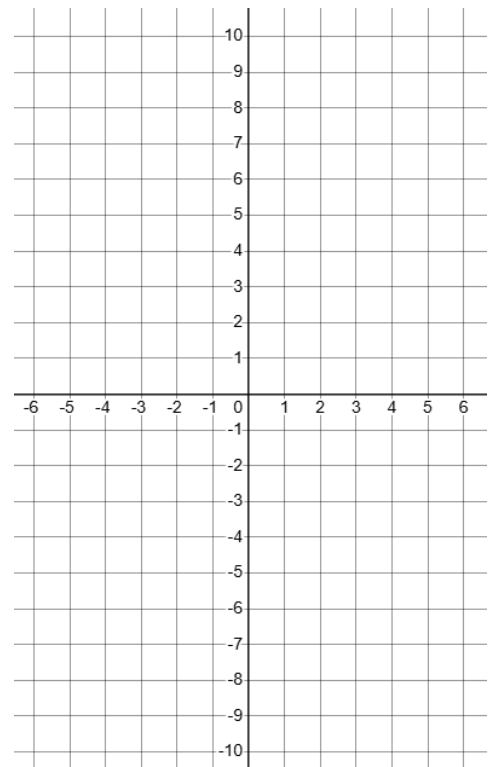
Range:

Example 2: Graph the function by finding the vertex, x and y intercepts.

$$h(x) = -2x^2 + 4x + 6$$

Vertex

x -intercept(s)

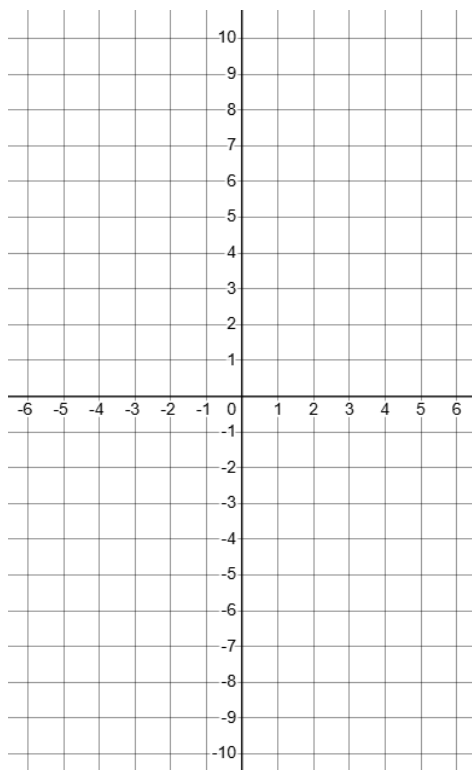


y -intercept

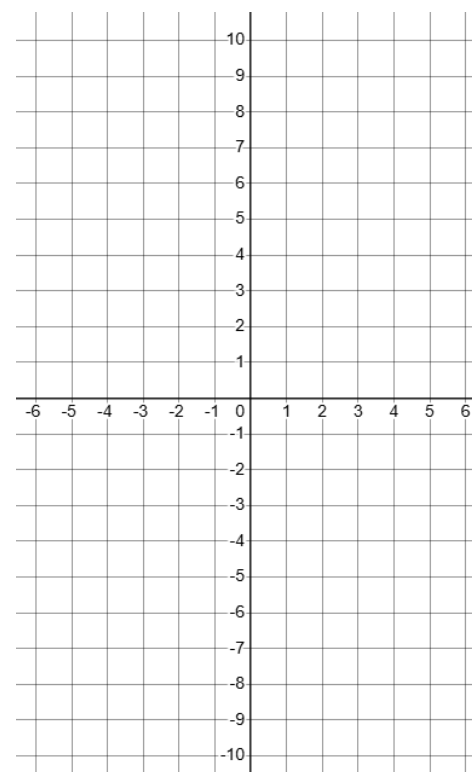
Domain:

Range:

Example 3: Graph the function $g(x) = 2x^2 - 8x$



Example 4: Graph the function $f(x) = -\frac{1}{2}(x + 1)^2 + 8$



Example 5: Given $f(x) = x^2 - 6x + 4$ and $g(x) = -2x + 25$

A. $f(-7)$

B. $g(-7)$

C. $f(x + 5)$

D. $g(x + 5)$

E. Solve for x if $g(x) = 17$

F. Solve for x if $f(x) = g(x)$



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M 090 Worksheet Packet